

PHAN HUU KHA

Aspiring Backend / Systems Engineer

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PROFILE

Software Engineering Student at Sai Gon University with a dedicated focus on Backend Development and DevOps. Proactively expanding knowledge in Java/Spring Boot and databases, while self-hosting a personal homelab server to gain hands-on practice with Docker and CI/CD. Eager to secure a Backend / DevOps Internship to learn production workflows from senior engineers and actively contribute to the team.

EDUCATION

Sai Gon University

Engineer's Degree in Software Engineering

Sept. 2023 – June 2028

Current GPA: 3.4/4.0

TECHNICAL SKILLS

Languages: Java (17/21 LTS), SQL, JavaScript, Bash Shell

Backend Technologies: Spring Boot 3, Spring Security 6, Spring Data JPA/Hibernate, WebSocket (STOMP), RESTful APIs

Databases, Caching & Queues: PostgreSQL, MongoDB, Redis, Apache Kafka (KRaft, Avro)

DevOps & Observability: Docker, Jenkins Pipeline, Tailscale (Funnel), GitHub Actions, Nginx, Prometheus, Grafana, ELK

Testing & Architecture: JUnit 5, Mockito, Event-Driven Architecture, Distributed Systems, Write-Behind Caching

PROJECTS

ChatWeb – Real-Time Distributed Messaging Platform

[Code](#)

Role: Backend Developer (Team Project – Backend Owner) | Java 21, Spring Boot 3.5, Kafka, Redis, PostgreSQL, MongoDB, WebSocket, Nginx, Docker

- Solely developed the entire backend system, defining RESTful APIs and STOMP WebSocket contracts for frontend integration.
- Researched and implemented distributed WebSocket messaging across multi-instance nodes using Redis Pub/Sub, avoiding sticky sessions.
- Built an asynchronous event-driven pipeline via Apache Kafka (KRaft) with compact Apache Avro schemas and automated DLT retries.
- Applied an asynchronous write-behind persistence pattern via Kafka batch consumers and MongoDB BulkOperations to cut disk I/O.
- Implemented secure authentication with Spring Security 6 and stateless JWT (access/refresh tokens via HttpOnly cookies) and RBAC.
- Configured Nginx reverse proxy (SSL/TLS, rate limiting, upstream TLS) and automated CI/CD via GitHub Actions.

Confession-Wall – Homelab Server Infrastructure & Automated Jenkins CI/CD

[Live](#) | [Code](#)

Role: Homelab Infrastructure & DevOps (Personal Lab) | Bare-Metal Linux, Tailscale & Funnel, Jenkins CI/CD, Docker Compose, Nginx

- Provisioned and self-hosted a dedicated Jenkins CI/CD service (:8443) and containerized workloads on a bare-metal Linux server.
- Configured zero-trust remote access via **Tailscale** and **Tailscale Funnel**, safely routing public HTTPS web endpoints without exposing home router ports or dealing with CGNAT.
- Constructed an end-to-end 5-stage Declarative Jenkins Pipeline triggered via GitHub webhooks, automating in-memory H2 Maven testing, multi-stage Docker builds, and Docker Hub image tagging.
- Automated remote host deployment via SSH, executing zero-downtime rolling container updates with an automated 10-iteration curl health-check polling loop and automated failure rollback logging.

LANGUAGES

Vietnamese: Native

English: Reading documentations and basic communication